ALRS v2: Unified ProjectState Architecture, 4-Quadrant Topology & Blackboard Synchronization

Root Schema: ProjectState (Pydantic v2) | UUID Session Context | Strict Sequential Stage Lifecycle Gating

1. FIVE SEQUENTIAL WORKFLOW STAGES (Sequential Lifecycle Gating)

Stage 1: Research

stage_1_research

- Seed Paper Intake & RQs
- OpenAlex API Discovery
- 1-Hop GraphSelect BFS
- Paper Library Metadata

transition_to_stage (Papers & Graph)

Stage 2: Analysis

stage_2_analysis

- 768-dim Gemini MRL Embeddings
- Cosine Similarity Matrix
- 3-Tier Centrality (GDS/NX)
- Gap Deficit Detection

transition_to_stage (Gaps & Rankings)

Stage 3: Planning

stage_3_planning

- Thematic / Chronological Arc
- Section Blueprints Synthesis
- Mandatory DOI Allocations (≥3)
- Target Word Counts

transition_to_stage (Outline & Blueprints)

Stage 4: Writing

stage_4_writing

- Closed-Loop Drafting Loop
- 5-Dimension Rubric Critique
- GCDL Zero Hallucination Gate
- Full Markdown Synthesis

transition_to_stage (Approved Drafts)

Stage 5: Export

stage_5_export

- IEEE / ACM / Article Templates
- Deterministic BibTeX Clean
- Headless pdflatex Compile
- GraphML Network Export

Each Stage Contains 4 Uniform Quadrants

2. UNIFORM 4-QUADRANT SUBSTATE MODEL (Replicated Across All 5 Stages)

Quadrant 1 (Q1): META

- **Schema:** StageMeta
- **status:** not_started | in_progress | completed | failed
- **agent_name:** Active execution agent tag
- **timestamps:** started_at, completed_at, updated_at
- **telemetry:** progress_percentage, iteration_count, error_message

Quadrant 2 (Q2): INPUTS

- **Schema:** Stage-Specific Pydantic Inputs
- **Parameters:** target_level ('master' | 'phd'), word_count
- **Thresholds:** similarity_threshold, max_depth, max_papers
- **Weights:** similarity_weight (0.70), pagerank_weight (0.30)
- **Directives:** active research questions, seed DOIs, style

Quadrant 3 (Q3): ARTIFACTS

- **Schema:** Concrete Produced Deliverables
- **Stage 1:** papers (dict[doi, PaperMetadata]), citation_edges
- **Stage 2:** similarity_matrix, clusters, gaps, rankings
- **Stage 3:** outline (OutlineStructure), section_blueprints
- **Stage 4:** section_drafts, section_reviews, full_draft_markdown
- **Stage 5:** tex_path, bib_path, compiled PDF, graphml

Quadrant 4 (Q4): NOTES

- **Schema:** list[StageNote]
- **Fields:** id, timestamp, author, category, content, tags
- **Agent Reflections:** Qualitative caveats & heuristics
- **Trade-off Logs:** Traversal pruning rationale, cluster limits
- **Contextual Memory:** Preserved across LLM turns

Selective Dot-Path Navigation: `ProjectState.get_by_dot_path('stage_1_research.artifacts.papers.10.1145/3292500.title')` ► Prevents Context Overflow

Cross-Cutting Insights Dispatched to Shared Board

3. PUBLIC MESSAGE BOARD: ASYNCHRONOUS BLACKBOARD COMMUNICATION

message_board: list[BoardMessage] (Persistent Cross-Agent Communication Channel)

💡 **insight**	🍀 **serendipity**	🕳 **gap**	⚠️ **warning**	⚖️ **decision**	🧪 **hypothesis**
Cross-paper synthesis	Unexpected citation link	Literature deficit / RQ void	Contradictory findings	Scope / Narrative choice	Testable proposition

- **Payload Structure:** id (UUID), timestamp (UTC), author_agent, message_type, title, content (Markdown), stage, tags, related_paper_dois
- **DOI Grounding:** Strict automatic regex normalization (strip url, prefix, punct) guaranteeing canonical `10.xxxx/...` linkage
- **Acknowledgment Tracking:** acknowledged_by: list[str] vectors track subscriber awareness (e.g. ['PlannerAgent', 'WriterAgent'])
- **State Tools:** `post_to_message_board(type, title, content, dois) | read_message_board(type, stage, unacknowledged_only)`

State & Messages Persisted Atomically

4. TRI-LAYER PERSISTENCE ARCHITECTURE & RECOVERY HIERARCHY

Layer 3: Relational MySQL

Audit & Historical Storage

- `project_states` audit table
- JSON document column + versioning
- Async background write dispatch
- Disaster recovery fallback

2. Async DB Audit Write

Fallback Recovery

Layer 2: Atomic Disk JSON

Canonical Storage File

- `project_state.json` per project
- Atomic swap: `.tmp.{uuid}` fsync
- Zero risk of corrupted state writes
- Path traversal validation

1. Atomic Disk Commit

Session Hydration

Layer 1: ADK ToolContext

In-Memory Session State

- Zero-latency dictionary access
- `tool_context.state['project_state']`
- Live agent invocation scope
- Pre-model hydration hook